

A Two-level Memetic Path Planning Algorithm for Unmanned Air/Ground Vehicle Cooperative Detection Systems

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Abstract—The Unmanned Air/Ground Vehicle (UAV/UGV) cooperative detection systems are widely used in terrain reconnaissance, disaster rescue and urban/rural buildings survey, in which the UAV in the air can provide a broad view and take photography, while the UGV on the ground can offer sufficient load capacity and serve as repeater stations as well. The path planning of UAV/UGV is one of the most important issues in the cooperative detection system, and its studies have attracted great attentions in recent years. In this study, a two-level memetic path planning algorithm is proposed to help the unmanned air/ground vehicle cooperative systems detect realistic urban illegal building, considering the limits of the endurance capacity of the UAV, the load capacity of the UGV, and the communication restrictions between them. Experiments on both the synthetic data and the realistic urban illegal building data show the effectiveness of the proposed algorithm on solving the path planning problem in the UAV/UGV cooperative system for urban illegal building detection tasks.

Unmanned Air/Ground Vehicle, cooperative detection systems, path planning, memetic algorithm

I. INTRODUCTION

In recent years, the studies on the Unmanned Air/Ground Vehicle (UAV/UGV) cooperative detection systems have attracted great attentions due to their wide applications on both military and civilian fields, such as terrain reconnaissance [1]–[3], border patrol [4], [5], target tracking [6], [7] and disaster rescue [8], [9]. The UAVs in the air have a fast flight speed and broad vision, which can quickly negotiate obstacles and traverse multiple locations. The UGVs on the ground have a good endurance capability and stability, which can load supplies and travel a long distance [10]. The cooperation of UAVs and UGVs can comprehensively realize their advantages on the realistic complicated tasks in terms of speed, vision, capability and stability.

The UAV/UGV cooperative systems [9], [11] consist of independent cooperative unmanned aerial vehicles (UAVs), unmanned ground vehicles (UGVs) and the control center, aiming to coordinate the UAVs and UGVs to complete realistic tasks while maintaining the communication restrictions between the vehicles and the control center [12], [13]. For instance, in a task of disaster relief, UAVs obtain the geomorphic information and send the aerial map to the ground monitoring center, while UGVs provide relief supplies to the disaster area based on the aerial map and the path planning generated by the ground monitoring center [9].

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The path planning is one of the most common but important issues in UAV/UGV cooperative systems for real tasks in the presence of the limits of the endurance capacity of the UAV, the load capacity of the UGV, the obstacles and the communication restrictions [14]–[16]. Most of the existing works mainly focus on the path planning of either UAVs or UGVs, see the comprehensive review in [17]. Recently, some works have been proposed for the path planning in both UAVs and UGVs in real tasks. For instance, in the target tracking mission, Yu et al. [6] considered the path planning as a probability maximization problem by modeling the moving target state as the dynamic occupancy grid. In the reconnaissance mission, Manyam et al. [18] developed a mixed integer linear programming (MILP) algorithm with a branch-and-cut strategy to plan the paths for the UAV and UGV, considering the communication constraints between the aerial and ground vehicles. In the autonomous package delivery tasks, Mathew et al. [19] solved a shortest path planning problem under the limitations that the truck can only drive on the existing street and the quadrotor should visit all the given locations to deliver packages. Actually, the path planning problem of UAV/UGV cooperative systems can be generalized to the Travelling Salesman Problem (TSP) [20] in the presence of some limits for real tasks, and it can be solved by some classic heuristic algorithms, such as the ant colony algorithm [21], the simulated annealing algorithm [22], the genetic algorithm [23] and the memetic algorithm [24].

In this paper, we study the path planning of the UAV/UGV cooperative systems with camera for the detection of non-standard buildings in cities. The UAV has the advantages in speed and vision, while the UGV has the advantages in capability and stability. Moreover, the UAV is generally limited by its energy and load capability, while the UGV is limited by its speed and vision. In a city, the buildings that need to be detected are widely distributed into a large area, and their non-standard objects are located in different heights. Although some works about the path planning of UAV or UGV detection systems has been proposed, the path planning of UAV/UGV cooperative systems remains to be unsolved. Moreover, the traditional UAV or UGV detection systems cannot work well for the detection of non-standard buildings in cities due to their limits.

To detect non-standard buildings in cities, we propose a UAV/GAV cooperative detection system. In this system, the UGV can carry the UAV and meanwhile charge the drone automatically, while the UAV can take a photo for the non-standard buildings from an aerial perspective. Moreover, we

propose a two-level memetic path planning algorithm (called as Two-MA) to help the UAV/GAV system detect realistic urban illegal building, considering the limits of the endurance capacity of the UAV, the load capacity of the UGV, and the communication restrictions between them. Two-MA first adopts a K-means algorithm to divide the buildings in cities into clusters, in which the buildings in the same clusters are closer than those in different clusters. Then, Two-MA uses the concept of hierarchical path planning, in which the UGV executes the global path planning across different clusters while the UAV executes the local path planning of buildings in the same clusters. Experiments on both the simulated data and the real data demonstrate the effectiveness of the two-MA algorithm.

II. RELATED BACKGROUNDS

A. The UAV/UGV cooperative system

The UAV/UGV cooperative system consists of UAVs, UGVs and a control center. The UAVs perform aerial photography in the air, while the UGVs execute loading and transportation on the ground. Moreover, the UGV can serve as the take off and landing for the UAV. The control center executes the compute-intensive tasks such as data transmission, data processing, task scheduling and vehicles coordinating. For instance, in the intelligence surveillance and reconnaissance (ISR) mission, the control center first plans the path and dispatchs asynchronous tasks for the UAV and UGV, then processes the data transmitted from UAV/UGV to give further instructions. A comprehensive survey for the UAV/UGV cooperative system is shown in [25]. An experimental cooperative UAV/UGV cooperative system is shown in Fig. 1.



Fig. 1: A real UAV/UGV cooperative system for experiments.

B. Path planning in UAV/UGV cooperative systems

Here, the path planning problem in UAV/UGV cooperative system aims to find the shortest path from a given position to a destination position which passes through all buildings. This problem can be modeled as a typical travelling salesman problem (TSP) [20] which is a classical NP-hard problem.

Given n possible positions, the path planning problem aims to find a possible path G across the n positions, so as to minimize the distance of the paths. It can be mathematically described as follows.

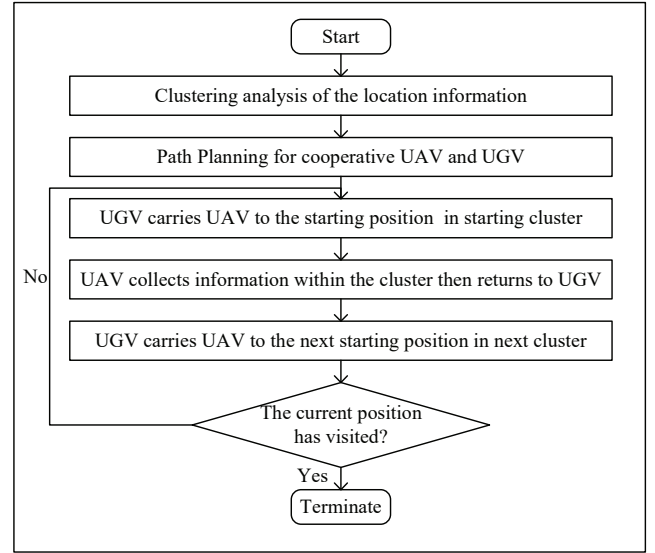


Fig. 2: Framework of the system model.

$$TSP(G) = \min \left(\sum_{i=1}^{N-1} D_i D_{i+1} + D_N D_1 \right), \quad (1)$$

in which $D_i D_{i+1}$ represents the distance from a build i to a building $i + 1$ in cities for the task of detection of non-standard buildings.

III. MODELS AND METHODS

In this part, we introduce the heterogeneous multirobot team with cooperative UAV and UGV to complete tasks of detection of non-standard buildings in cities.

The whole framework of the cooperative system is shown in Fig. 2. Generally, the operations of the systems consists of three steps: clustering analysis, path planning for the UAV and the path planning for the UGV.

Step 1) Clustering analysis. Generally, the buildings in a cite are widely distributed into a large area, and their non-standard objects are located in different heights. To quickly find a good path, we first divide the areas with buildings into a set of clusters, in which the buildings in the same cluster are close whereas those in different clusters are far. Here, we choose the K-means clustering algorithm [26] to generate clusters for buildings based on the Global Position System (GPS) data. The reason for choose K-mean clustering algorithm is that it can can achieve a good tradeoff between the clustering effect and the time complexity.

Considering the limits of the endurance capacity of the UAV, the load capacity of the UGV, and the communication restrictions between them, we use the UAV to gather image information of the buildings for further analyses within the cluster, and use and the UGV to carry the UAV to traverse different clusters.

Step 2) Path planning for the UAV.

The UAV within a cluster starts from the center of a cluster, and then passes through the rest of positions. Finally, it returns to the center of the cluster. Therefore, the path planing of the UAV can be modeled as a traveling salesman problem (TSP) from the node novel, which needs to find a shortest path that travels all nodes in a cluster. To solve this TSP problem, we propose a memetic algorithm which is a combination of a genetic algorithm [23] and a local search.

Step 3) Path planning for the UGV. The UGV aims to carry the UAV across different clusters. In this case, the path planning for the UGV across different clusters can be modeled as the TSP problem for the cluster level. Specifically, the path planning problem for the UGV aims to find a shortest path that travels the centers of all clusters. To solve the TSP problem in the cluster level, we adopt the proposed memetic algorithm for the path planning of UAV when we set each cluster as a super node.

Step 4) Geographic information collection. Based on the planned paths for the UAV and UGV, the heterogeneous cooperative system collects the geographic information of the buildings as follows. First, the UVA carries the UAV to the center of a cluster. Then, the UAV collects the information of the buildings within the cluster based on the UAV path. Next, the UGV carries the UAV to the next cluster based on the UGV path. The steps above are executed until all information of the buildings are collected.

A graphical representation of the whole system is given in Fig. 3.

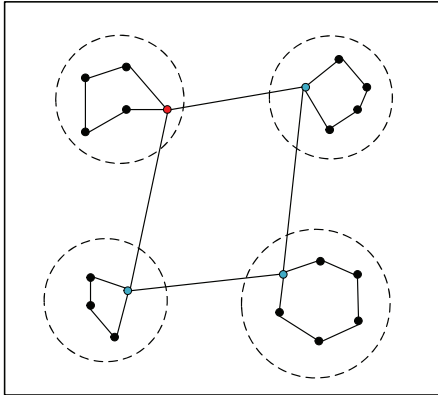


Fig. 3: Graphical representation of the system model. The dotted circles represent different clusters, and the red/blue nodes represent the start position of the UAV path in clusters.

In the following, the K-means clustering algorithm and the proposed memetic algorithm are described.

A. K-means clustering algorithm

Many clustering algorithm can be used to find the clusters of buildings in cites, including K-Means clustering [26], Mean-Shift clustering [27], Density-Based Spatial Clustering [28] and Hierarchical clustering [29]. Here the K-Means clustering algorithm is adopted due to its low computational complexity and good clustering effect. The urban buildings

Algorithm 1 K-Means clustering algorithm

- 1: **Input:** The number of clusters: k ; The maximum number of iteration: I_{max} ; The original dataset: G .
- 2: $\mathbf{E} \leftarrow \text{InitializeCentroids}(G, k)$;
- 3: **repeat**
- 4: $\mathbf{C} \leftarrow \text{PartitionClusters}(\mathbf{E})$;
- 5: $\mathbf{E}_{new} \leftarrow \text{RecomputationCentroids}(G, \mathbf{C})$;
- 6: $\mathbf{E} \leftarrow \text{UpdateCentroids}(G, \mathbf{E}_{new})$;
- 7: **until** $\text{TerminationCriterion}(I_{max})$
- 8: **Output:** The k centroids $\mathbf{E}$ and cluster divisions $\mathbf{C}$.

Algorithm 2 Framework of Two-MA for path planing

- 1: **Input:** Position indexes: C ; Population size: S_{pop} ; Maximum number of generations: G_{max} ; Crossover probability: P_c ; Mutation probability: P_m ; The size of mating pool: N_m ; the tournament size: N_t ;
- 2: $\mathbf{P} \leftarrow \text{Initialization}(C, S_{pop})$;
- 3: **repeat**
- 4: $\mathbf{P}_{parent} \leftarrow \text{Selection}(\mathbf{P}, N_m, N_t)$;
- 5: $\mathbf{P}_{child} \leftarrow \text{GeneticOperation}(\mathbf{P}_{parent}, P_c, P_m, N_t)$;
- 6: $\mathbf{P}_{new} \leftarrow \text{LocalSearch}(\mathbf{P}_{child})$;
- 7: $\mathbf{P} \leftarrow \text{UpdatePopulation}(\mathbf{P}_{new})$;
- 8: **until** The current iteration reaches G_{max} .
- 9: **Output:** Convert the fittest chromosome in $\mathbf{P}$ into the path of UAV/UGV.

data are collected by the Global Positioning System (GPS), containing two dimensions representing the longitude and latitude of the buildings, respectively. The K-Means clustering algorithm can divide the original data into k different clusters based on the k initial centroids, where k is predefined number of clusters. The framework of K-Means clustering algorithm is given as **Algorithm 1**.

In **Algorithm 1**, the $\text{InitializeCentroids}()$ procedure randomly selects k nodes in the dataset as the initial centroids $\mathbf{E}$, then the $\text{PartitionClusters}()$ operation divides the nodes into k clusters $\mathbf{C}$ based on the initial centroids $\mathbf{E}$. The nodes that partitioned into the same cluster share the nearest mean. The $\text{RecomputationCentroids}()$ function computes new centroids $\mathbf{E}_{new}$ in the current clustering division $\mathbf{C}$. The whole algorithm is terminated when the iteration reaches the I_{max} or the centroids $\mathbf{E}$ are no longer changed.

B. Two-MA for path planning for the UAV and UGV

Two-MA aims to minimize the objective in Eq. (1) (minimize the distance of paths) for the UAV and UGV by simultaneously optimizing a population of solutions. The framework of Two-MA for path planing is given as **Algorithm 2**, and the corresponding operations are detailed next.

1) *Initialization:* In the initialization step, the $\text{Initialization}()$ function randomly generates the initial population with S_{pop} individuals, in which each individual C represent a possible planning path. Specifically, the position indexes C are a set of positions' indexes $\{C_i \in 1, 2, \dots, N\}_N$, and the planning path passes through all buildings from

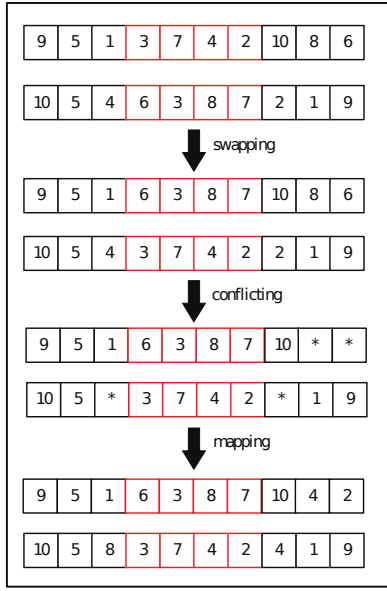


Fig. 4: Schematic illustration of the crossover operation. Here, the generated values r_1 and r_2 are 4 and 7, respectively.

the first position to the final position. For instance, the set $\{1,2,3,4,5,6,7,8,9,10\}$ represents 10 different positions, and the described path starts from the building 1, and ends with the building 10.

2) *Selection*: The aim of the selection step is to preserve individuals with high fitness to the next generation. In this research, the typical tournament selection strategy is adopted, which randomly selects N_t chromosomes from the initial population $\mathbf{P}$ and preserves the best one into the mating pool for N_m times. The tournament selection strategy guarantees the diversity and superiority of the population.

3) *GeneticOperation*: The GeneticOperation contains the crossover and mutation operations to simulate the evolution of creatures in the nature. The crossover operation first generates two random numbers r_1 and r_2 , which range from 1 to N , and then swaps the selected part of two parent chromosomes from r_1 and r_2 . The crossover operation may introduce duplicate codings to the original chromosome. In this case, we conserve the swapped chromosome segments, and map the duplicate codes in other segments to the corresponding codes of the conflicting ones in the swapped segments. Fig. 4 shows a detailed crossover process of two chromosomes. Similarly, in the mutation operation, we first randomly generate two numbers, r_1 and r_2 , which range from 1 to N , then swap the corresponding code. A schematic illustration of the mutation operation is shown in Fig. 5.

4) *LocalSearch*: To improve the performance of the genetic algorithm, we introduce the evolution reverse operation into the local search. Specifically, for a chromosome, first, some genes in a chromosome are randomly chosen. Then, their gene values are reversed. Next, if the reversion can improve the fineness of the chromosome, then we accept this reversion. Otherwise, we set the chromosome as its original

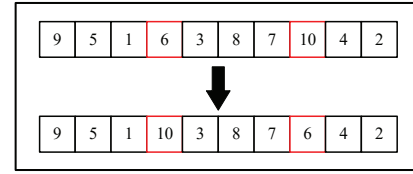


Fig. 5: Schematic illustration of the mutation operation. Here, the generated values r_1 and r_2 are 5 and 7, respectively.

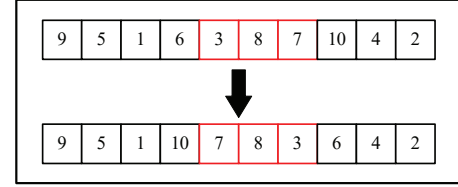


Fig. 6: Schematic illustration of the evolution reverse operation. Given two randomly generated integers $r_1 = 5$ and $r_2 = 7$, we reverse the corresponding segment of the original chromosome when this reversion can improve the fitness of the chromosome.

state. A schematic illustration of the local search is shown in Fig. 6.

5) *UpdatePopulation*: In this step, we choose a population of chromosomes with high fitness values from the original population and the generated chromosomes as the the population for the next generation. This step guarantees the evolve of the population at each iteration.

IV. EXPERIMENTAL RESULTS

In this section, we test the proposed algorithm (Two-MA) on both the synthetic data set and the real data to demonstrate its effectiveness. Moreover, we compare Two-MA with **GA** and **Two-GA** which are the traditional genetic algorithm for TSP without the local search process. Here, **GA** only considers the path planning of a UAV, while **Two-GA** gives the path planing of both the UAV and UGV. All results are averaged over 30 independent trials under the parameters setting in Table I.

A. Synthetic data set

The synthetic data set is composed of 1,000 nodes and 25 clusters. Each cluster has 40 nodes randomly generated in a given range in a coordinate axis. Fig. 7 shows the representation of the synthetic data. The test results of all algorithms

TABLE I: Setting of parameters of all algorithms.

Parameter	Meaning	Value
G_{max}	The number of iterations	200
S_{pop}	Population size	100
N_m	Size of the mating pool	$\frac{S_{pop}}{2}$
N_t	Tournament size	2
P_c	Crossover probability	0.9
P_m	Mutation probability	0.05

TABLE II: Results of all algorithms about the path length on the synthetic data.

Path Length	UAV	UGV	Total
GA	1223504460.8908	0	1223504460.8908
Two-GA	849.73	159.28	1009.01
Two-MA	547.18	128.72	702.90

are given in Table II. The results show that **Two-MA** can obtain the shortest path. Specifically, its obtained path is 702.9 lower than that of GA (1223504460) and Two-GA (1009). Moreover, The single path planning algorithm (GA) has a extremely longer path than the two-level path planning algorithms (Two-MA and Two-GA), which validates the effectiveness of the clustering strategy and the UAV/UGV cooperative system. To further show the effectiveness of Two-MA, we show the planned path of Two-MA on the synthetic data in Fig. 8. The results show that Two-MA can effectively find the shortest paths among buildings in both the same clusters and different clusters.

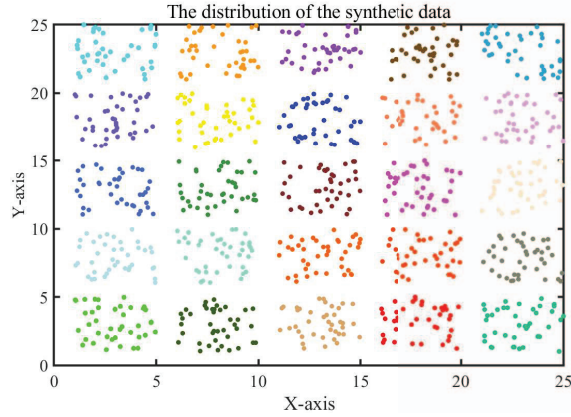


Fig. 7: Representation of the synthetic data in a coordinate axis. Here, nodes with the same colors are in the same clusters.

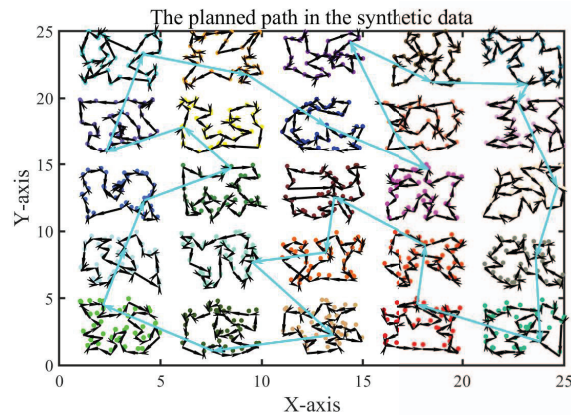


Fig. 8: The planned path of UAV/UGV cooperative system in the synthetic data set. The black arrows represent the path within a cluster, while the blue arrows represent the UGV path across different clusters.

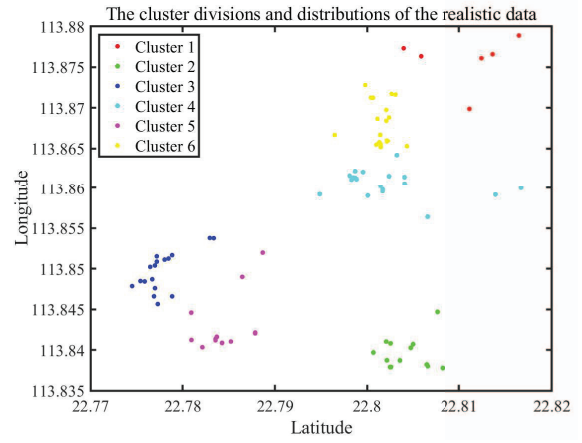


Fig. 9: The cluster divisions and the distributions of the realistic illegal buildings data. The solid dots with different colors represent the data that partitioned into different clusters. In the figure there are 6 clusters.

TABLE III: Results of all algorithms about the path length on the real data.

Path Length	UAV	UGV	Total
GA	75.25	0	75.25
Two-GA	22.95	12.20	35.15
Two-MA	21.95	11.60	33.55

B. Real-world data set

The real-world data set is extracted from the illegal buildings distributions from Yanluo Street, Shenzheng City, Guangdong Province, China, which consists of 98 GPS (Global Positioning System) points with exact latitude and longitude data of the illegal buildings. Fig. 9 shows the distributions of the realistic data, in which the horizontal/vertical axis represents the latitude/longitude data. This data set can be divided into 6 clusters by the K-means algorithm.

Fig. 10 shows the path planning of Two-MA for the UAV and UGV cooperative system on the real data. The path lengths for the UAV inside the six clusters are 34.31 kilometers, 37.06 kilometers, 58.25 kilometers, 29.43 kilometers, 33.00 kilometers and 27.50 kilometers, respectively, while the path length for the UGV is 115.97 kilometers. Those indicate that the path inside a cluster is much shorter than that across different clusters. In this case, we can use the UAV to detect non-standard buildings in the same cluster while use the UGV to take the UAV across different clusters.

Table III provides the comparison results of all algorithms about the path length on the real data. The results further validate that the two-level path planning algorithm (Two-GA and Two-MA) for the UAV/UGV cooperative system can achieve much shorter path than the single-level path planning algorithm (GA). Moreover, it also show that our algorithm (Two-MA) can obtain the shortest path, which validates the effectiveness of Two-MA.

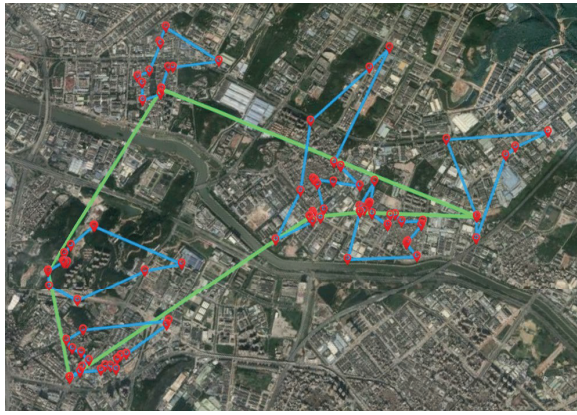


Fig. 10: The planned paths in the realistic environment. The green (blue) line represents the path of the UGV (UAV).

V. CONCLUSIONS

In this paper, we proposed a UAV/UGV cooperative detection system for the detection of non-standard buildings in cities, in which the UAV and UGV operate alternately under the consideration of UAV's energy and UGV's loading capacity. Moreover, we proposed a two-level memetic path planning algorithm (Two-MA) to find the shortest paths among buildings from both the node levels and the cluster levels. In addition, experimental results on the synthetic data as well as the real data proved the superiority of Two-MA over the classical algorithms (GA and Two-GA) on finding the shortest paths for the detection of non-standard buildings in cities.

In future work, we will study the multiple UAVs/UGVs cooperative detection system for the detection of non-standard buildings in cities under more real constraints.

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